

Design & Analysis of Algorithms

Lab 1: Algorithm Performance Analysis Workbench

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Q1. 1.4 Algorithm Execution Observation Table (5 marks)

```
import math
def recursive_fibonacci_count(n):
    if n<=1:
        return 1
    return 1+recursive_fibonacci_count(n-1)+recursive_fibonacci_count(n-2)
def generate_execution_observation_table(sizes):
    result=[]
    result.append("Algorithm Execution Observation Table")
    result.append("InputSize RecursiveFactorial IterativeFactorial RecursiveFibonacci
IterativeFibonacci LinearSearch BinarySearch BubbleSort InsertionSort")
    for n in sizes:
        recursive_factorial = n+1
        iterative_factorial = n
        recursice_fibonacci = recursive_fibonacci_count(n)
        iterative_fibonacci = n
        linear_search = n
        binary_search = math.floor(math.log2(n))+1
        bubble_sort = n*(n-1)//2
        insertion_sort = n*(n-1)//2
        result.append(f"{n} {recursive_factorial} {iterative_factorial} {recursice_fibonacci}
{iterative_fibonacci} {linear_search} {binary_search} {bubble_sort} {insertion_sort}")
    return result
```